Atrial fibrillation associated with chocolate intake abuse and chronic salbutamol inhalation abuse.

The use of substances as the substrate for atrial fibrillation is not frequently recognized. Chocolate is derived from the roasted seeds of the plant theobroma cacao and its components are the methylxanthine alkaloids theobromine and caffeine. Caffeine is a methylxanthine whose primary biological effect is the competitive antagonism of the adenosine receptor. Normal consumption of caffeine was not associated with risk of atrial fibrillation or flutter. Sympathomimetic effects, due to circulating catecholamines cause the cardiac manifestations of caffeine overdose toxicity, produce tachyarrhythmias such as supraventricular tachycardia, atrial fibrillation, ventricular tachycardia, and ventricular fibrillation.The commonly used doses of inhaled or nebulized salbutamol induced no acute myocardial ischaemia, arrhythmias or changes in heart rate variability in patients with coronary artery disease and clinically stable asthma or chronic obstructive pulmonary disease. Two-week salbutamol treatment shifts the cardiovascular autonomic regulation to a new level characterized by greater sympathetic responsiveness and slight beta2-receptor tolerance. We present a case of atrial fibrillation associated with chocolate intake abuse in a 19-year-old Italian woman with chronic salbutamol inhalation abuse. This case focuses attention on chocolate intake abuse associated with chronic salbutamol abuse as the substrate for atrial fibrillation.